Future work: testing the open boundaries

The staged research program follows directly from the boundaries in sec. 11.13. It separates public-library reproducibility, server theory, portable source-protocol work, task generalization,

Future work: testing the... (part 2)

and deployment validation so that success in one lane cannot silently certify another.

Future work: testing the... (part 3)

The order is deliberate. Simulation-study guidance recommends declaring the estimand, data-generating mechanism, and Monte Carlo precision before treating replication as evidence (Morris et al. 2019),

Future work: testing the... (part 4)

while Monte Carlo error should be reported separately from an interval or a hypothesis test (Koehler et al. 2009).

Future work: testing the… (part 5)

For nested agents, trials, and seeds, the resampling unit must respect the dependence structure (Loy and Korobova 2021). Accordingly, the phase plan records a primary unit and a falsifier for each extension; a larger sample or more elaborate diagram is not itself a stronger claim.

Future work: testing the... (part 6)

The implementation registry also separates smoke, pilot, and confirmatory profiles. Pilot worlds select budgets and calibration settings but never enter confirmatory intervals or headline values.

Future work: testing the... (part 7)

Each completed run must bind its source bundle, configuration, dataset bytes, device, checkpoints, outputs, and completion status into a verifiable receipt. A negative or null scientific result remains a valid citable outcome when those implementation and provenance gates pass; it blocks the intended positive claim,

Future work: testing the... (part 8)

not the software release.

Future work: testing the... (part 9)

A separate Friston protocol lane will resolve a machine-readable parity matrix before reconstructing source experiments in Python. Until every required source parameter, routine, unit, and estimand is recovered, that lane will be described as a paper-constrained reconstruction rather than an exact replication.

Make the sharp server heuristic variational

The asymmetry between the robustness axes (sec. 6.3) motivates a consequential theoretical design question that is explicitly parked outside the authorized roadmap.

Make the sharp server… (part 2)

The client-side β/rcce update carries a derived, loss-specific bounded-influence result under the matching assumptions; the sharp server-side `robust_aggregate` carries only its recovery limit (eq. 10).

Make the sharp server… (part 3)

The new variational aggregator (sec. 8.8.2, sec. 15) supplies a *rigorous* server-side rule — exact block updates descending the stated free energy eq. 12 through non-increasing block updates,

Make the sharp server... (part 4)

with a proven raw effective-weight bound and the same recovery corner. What it costs is conservatism: it is the maximum-entropy consensus, not the sharp accuracy-maximizer.

Make the sharp server... (part 5)

That parked question is sharper than before: write down a generalized variational objective in the FedGVI family (Mildner et al. 2025), informed by recent closed-form GVI characterizations (Nguyen and Westerhout 2026), logarithmic-pool weighting theory (Carvalho et al. 2023),

Make the sharp server... (part 6)

and robust divergence-weighted federated aggregation (Li et al. 2022), whose closed-form minimizer is competitive with the empirical reweighting across declared contamination regimes.

Make the sharp server… (part 7)

That would combine axis 3's effective-weight bound with axis 2's empirical sharpness in one server rule. The recovery corner and the variational objective together supply two boundary conditions any such unification must satisfy.

Make the sharp server... (part 8)

Any empirical choice of `robustness` or `entropy_weight` will be made on separate calibration worlds using a proper log score, then frozen before confirmatory evaluation. This guards against selecting an apparent leader with evaluation truth and preserves null or reversed confirmatory outcomes.

Make the sharp server... (part 9)

The comparison family will include logarithmic and linear pools, the current heuristic, the variational family, and a centered-log-ratio geometric-median control; none inherits a parameter-space robust-federated-learning guarantee merely by operating on beliefs.

Promote the baseline to original FedGVI scale

The bounded-influence result is anchored locally by the small NumPy logistic-regression baseline (fig. 21).

Promote the baseline to... (part 2)

Promoting it to the GPU-scale Bayesian-neural-network experiments of the source paper (Mildner et al. 2025) — the experiments deferred here (sec. 11.13) —

Promote the baseline to... (part 3)

would test whether the per-client robustness curve holds at the model capacity and contamination regimes where federated learning (McMahan et al. 2017) actually operates.

Promote the baseline to... (part 4)

It would connect the discrete-POMDP result to the partitioned-VI line (Ashman et al. 2022; Bui et al. 2018). The planned comparison would also require posterior-parameterization parity with Bayesian neural-network work, rather than treating the current deterministic point-mass MLP as a posterior (Mildner et al. 2025).

Promote the baseline to... (part 5)

The portable lane preserves the source protocol's site factors, client cavity, factor-replacement update in natural coordinates. A synthetic CPU/MPS pilot already exercises the cavity-conditioned local optimizer, explicit device and fallback receipts, and checkpoint/resume equivalence.

Promote the baseline to... (part 6)

It does not establish source-dataset parity.

Promote the baseline to... (part 7)

The next local campaign distinguishes a locked portable CPU/MPS profile from an exact source-scale CUDA profile that remains external until suitable hardware is available. FashionMNIST anchors source-dataset protocol parity, while MNIST and KMNIST test portability.

Promote the baseline to... (part 8)

A separate source-bound tabular pack will report proper-score effects per licensed dataset, with training-only preprocessing and byte-, split-, and license-level provenance; nested seeds will not be treated as independent datasets.

Extend hierarchical federation beyond the current stack

The governing caveat here is that added depth must be shown to *earn* generalization, not merely to execute:

Extend hierarchical... (part 2)

on the current sentinel task the deeper stacks match rather than beat the flat baseline on location (sec. 11.15). The extension therefore has two distinct fronts —

Extend hierarchical... (part 3)

carrying the recovery contract to deeper stacks, and finding a task family in which depth actually pays.

Extend hierarchical... (part 4)

The 2-level hierarchical POMDP (sec. 17.9) couples location inference to a single global context. The generic N-level architecture uses `build_nlevel_world` from the `fedference.pomdp` module and has already been exercised with a 3-level stack (sec. 17.11,

Extend hierarchical... (part 5)

sec. 17.12): a meta-context variable (L3) gates the context prior (L2) which in turn gates the location prior (L1).

Extend hierarchical... (part 6)

The empirical-prior top-down messages (eq. 38, eq. 39) remain valid variational steps at every depth, and the log-linear-pool federation at each level is bit-identical to the in-process result (Proposition 12).

Extend hierarchical... (part 7)

The natural next question is whether the limit-as-proof contract of sec. 11.13 survives still deeper hierarchies: does the recovery corner (context prior $\to$ uniform) remain checkable to machine precision when the L2 $\to$ L1 message is itself a function of an L3 belief coupled to an L4 belief,

Extend hierarchical... (part 8)

and can message-passing engines such as RxInfer (Bagaev et al. 2023) carry the generic alternating-minimization at scale?

Extend hierarchical... (part 9)

The current hierarchical diagnostic asks a narrower question. Under one configured surprise threshold, it classifies the non-gating top level below the threshold and the informative top level above it (sec. 18.1). This is an implementation control,

Extend hierarchical... (part 10)

not a posterior refit or an autonomous depth-selection result.

Extend hierarchical... (part 11)

A future breadth-and- depth search must define its candidate family, scoring rule, uncertainty, replication units, and out-of-sample evaluation before it can support a model- selection claim for the generic N-level stack.

Extend hierarchical... (part 12)

The next task family is deliberately controlled rather than merely deeper: partially observable Four Rooms and Key-Door will compare flat, oracle, learned, shuffled, and non-gating hierarchies at matched horizons and compute. Task is the higher-level replication unit. A gain in only one task will remain task-specific,

Extend hierarchical... (part 13)

and the larger campaign will not begin until the hybrid representation recovery gates pass.

Move from process transport to true multi-machine federation

The current deployment uses queue-backed single-machine processes and loopback sockets. Promotion to cross-host workers would address the remaining deployment caveat of sec. 11.13.

Move from process... (part 2)

That extension must preserve the bit-identical consensus property proved in Proposition 12.

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Move from process... (part 3)

The `federation/` package and federation tests already establish the API contract: a server collects serialized beliefs from 5 worker channels, fuses them with `robust_aggregate` at robustness $c = 1.5$, and broadcasts the consensus back over response channels, with bit-identity verified at True.

Move from process... (part 4)

The loopback socket path adds optional pre-shared-key frame integrity and file-backed digest-verified replay validation. The next systems step is an explicitly local Docker multi-node emulator with mTLS by default, HMAC compatibility, checkpoint/restart, and reproducible message-fault controls.

Move from process... (part 5)

It is not physical multi-host evidence.

That later claim requires receipts from distinct hosts, deployment-grade key management, timeout policy, and long-running orchestration across process restarts, but it does not require changing the mathematics.

Move from process... (part 6)

Secure aggregation, differential privacy, and Byzantine tolerance are separate future threat models; transport integrity alone would not establish any of them (Blanchard et al. 2017; Pillutla et al. 2022).

Move beyond categorical state spaces

This work is discrete-categorical, matching the community's worked POMDP (Da Costa et al. 2020). A first step is already in place: the closed-form Gaussian KL and Rényi divergences of sec. 16 show the divergence family — and its $\alpha \to 1$ recovery — carries over verbatim to Gaussian beliefs,

Move beyond categorical... (part 2)

scoped out of the categorical experiments.

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Move beyond categorical... (part 3)

Continuous-state Gaussian generative models would test whether the limit-as-proof contract survives the move off categorical state spaces — whether the recovery corner remains checkable to machine precision when the belief simplex is replaced by a Gaussian belief,

Move beyond categorical... (part 4)

and whether message-passing engines such as RxInfer (Bagaev et al. 2023) can carry the robust update at scale.

Move beyond categorical... (part 5)

Extending the structure-learning study (Smith et al. 2022; Friston and Penny 2011) into continuous models would, in parallel, test whether robust belief fusion and robust *structure* fusion compose.

Move beyond categorical... (part 6)

A minimal executable fixture now gates a discrete dynamics context over continuous position and velocity, Gaussian observations, and bounded actions. The bounded pilot now includes matched naive, robust, discrete-only, continuous-only, and oracle-context components, a singular-covariance rejection,

Move beyond categorical... (part 7)

and next-position predictive scoring.

Move beyond categorical... (part 8)

It is still a representation, recovery, and control surface rather than confirmatory task evidence. The full study must freeze calibration and budgets, use independently held-out worlds, retain outlier falsifiers, and execute the preregistered comparison before supporting a hybrid-task claim.